

CURSO ING2030

INVESTIGACIÓN, INNOVACIÓN Y EMPRENDIMIENTO

EMPRENDIMIENTO RESPONSABLE

Objetivo de la sesión

Clase 12

Reconocer aquellos elementos que componen una falta a la ética en los emprendimientos.

Comprender la importancia e impacto social que generan los emprendimientos

Nota por participación 18

Clase 12

- ¿Cuándo?: Durante el día de la clase 12.
- Tipo de participación: Asistencia
- Enunciado: De forma INDIVIDUAL, quienes asistan a la clase 12 podrán optar por Bonificación por Participación.
- Objetivo: Comprender, desde la ética y su rol en los proyectos, el impacto que tienen las soluciones que cada equipo está generando en la sociedad.

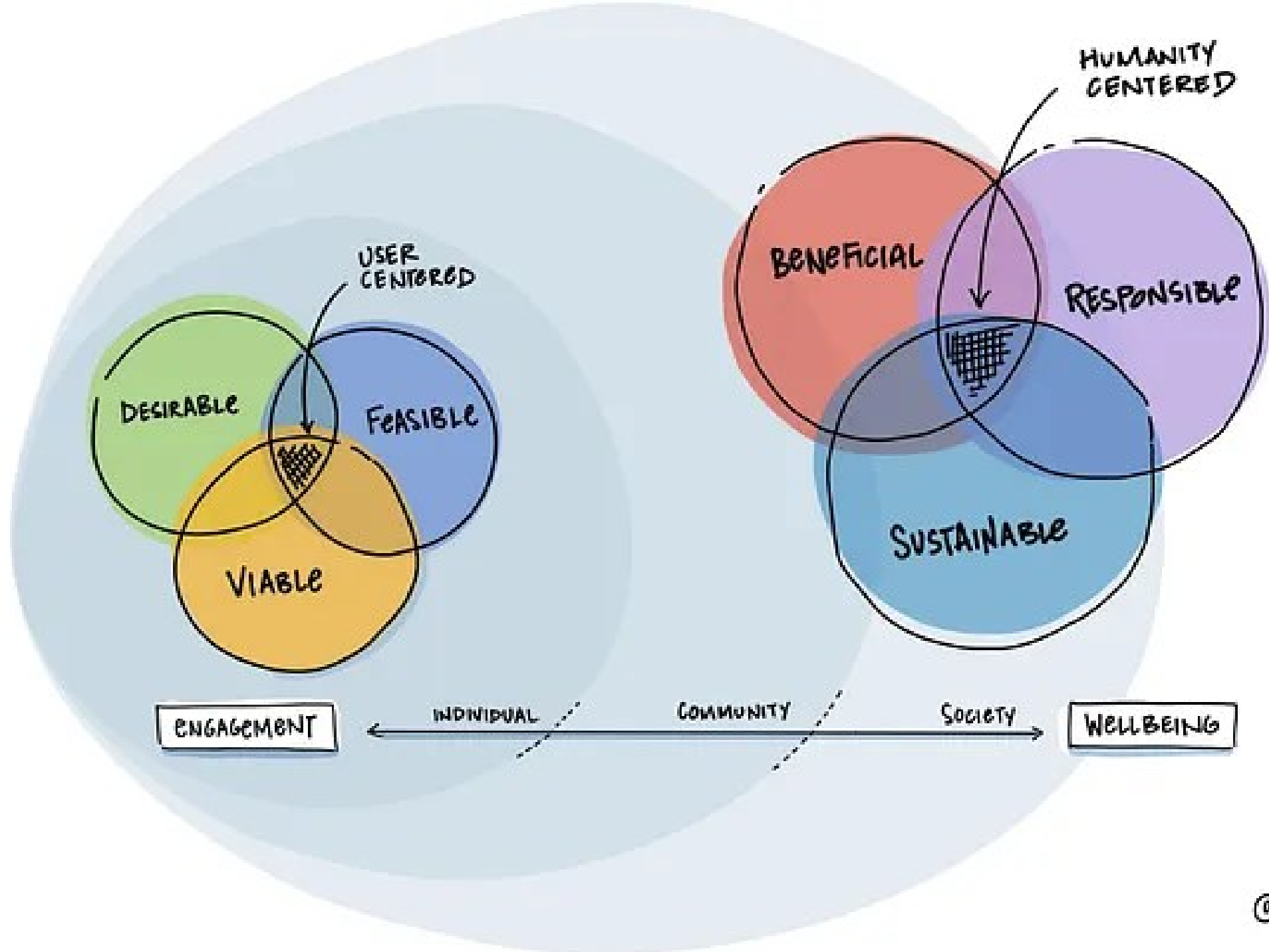


<https://www.youtube.com/watch?v=o4hkgELvs5M&feature=youtu.be>



<https://youtu.be/b9u9zPlh3V4?list=PL8FGQWmC19rO8qp-B14KFBDzJW TY3pqHD>

**¿POR QUÉ ENSEÑAR ÉTICA EN
EMPRENDIMIENTO?**



VENTUREBEAT.COM

Business

Feature

The ugly, unethical underside of Silicon Valley

ERIN GRIFFITH, FORTUNE DECEMBER 28, 2016 7:15 AM



Image Credit: Illustration by Alconic for Fortune

VB TRANSFORM

The AI event for business leaders

San Francisco
July 15 - 16

[Learn More](#)

Diciembre, 2016

ECORNER.STANDFORD.EDU

Innovation

Finding Our Values: A New Era of Entrepreneurship Education

Tom Byers, Stanford University | October 17, 2018



IN BRIEF: Entrepreneurship education is changing. Students must be equipped with the skills to make values-based decisions.

9 MINUTES

[+ add to collection](#) [♥ save for later](#)

Octubre, 2018

BUSINESSINSIDER.COM

The 18 biggest tech scandals of 2018

Paige Leskin and Nick Bastone Dec 29, 2018, 1:45 PM



Diciembre, 2018

PSMAG.COM

Pacific Standard



EMAIL NEWSLETTER
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OUR FREE
NEWSLETTER

NEWS IN BRIEF ECONOMICS EDUCATION ENVIRONMENT

HOME > IDEAS > SPECIAL PROJECTS > UNDERSTANDING GEN Z

ARE ELITE INSTITUTIONS TEACHING STUDENTS THE WRONG VALUES?

There's a clear need to rethink what "impact" means, given the concept's distorting effect on students' priorities and ethics.

Abril, 2019

ENTREPRENEUR.NYU.EDU

Entrepreneurial Institute

Ethical Entrepreneurship in Education

By Andy Moss Wed., May 8, 2019

An Active Landscape for Ethical Entrepreneurship Education

Entrepreneurship education is changing. The Omidyar Network, *Ethics in Entrepreneurship*, Zebras Unite and Stanford Innovation Lab proactively advocate ethics in entrepreneurship and offer some good practices of integrating ethics into entrepreneurship education. Even though they have different focuses, they all provide guidance to envision ethical challenges and build communities to empower ethical entrepreneurship. The NYU Entrepreneurial Institute is also developing materials to help young entrepreneurs to consider ethical risks from day one. By strengthening connections with people and organizations who have relevant experiences and insights, the Institute can foster a powerful network for ethical entrepreneurship.

Mayo, 2019

TELEGRAPH.CO.UK

Facebook removed from S&P list of ethical companies after data scandals



Save 3



Junio, 2019

“While current businesses impact the ethical standards used in present business dealings, **emerging entrepreneurial companies set the ethical tone for the future economic system of the world.**”

/ Bucar, Branko and Robert D. Hisrich (2001)

INNOVATION= INVENTION x COMMERCIALIZATION

/ BILL AULET – MIT

A STARTUP IS A COMPANY DESIGNED TO GROW FAST

/ PAUL GRAHAM – Y COMBINATOR

INNOVATION= INVENTION x COMMERCIALIZATION

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Estas dos definiciones abren una serie de dilemas éticos diferentes a los que enfrentan las empresas tradicionales

**ÉTICA RELACIONADA
AL EMPRENDIMIENTO**

**ÉTICA RELACIONADA
A LA TECNOLOGÍA**

What do we mean when we talk about ethical tech?

(..) it is critical to draw the **distinction between corporate and professional ethics**—ethics related to questions of business, professional conduct, humane treatment of workers, and/or corporate and social responsibility—**and ethics of technology**. What do we mean when we use the latter term?

The World Economic Forum argues that “technologies have a clear moral dimension—that is to say, a fundamental aspect that relates to values, ethics, and norms. Technologies reflect the interests, behaviors, and desires of their creators, and shape how the people using them can realize their potential, identities, relationships, and goals.”

Ethical tech is, at its heart, **a conversation focused on the relationship between technology and human values, the decisions we make toward technological advances, and the impacts they can have** (...) It is a multifaceted concept that can encompass a wide variety of issues, from data privacy to bias in algorithms, from replacing humans with machines to a commitment to not manipulating data or human responses.

ÉTICA RELACIONADA AL EMPRENDIMIENTO

THE DEFINITIVE RANKING OF THE RICHEST PEOPLE IN AMERICA

SPECIAL EDITION

Forbes



THE
FRESHMAN

ELIZABETH
HOLMES
LEADS THE
CLASS OF 2014

400
THE
FORBES

“What many consider to be entrepreneurial behavior is a set of actions fraught with ethical dilemmas. Entrepreneurs are often admired for the creative ways in which they overcome significant limitations, obstacles and sources of resistance to their new venture ideas.

Practices such as **bending or breaking rules**, putting **other people's resources at risk**, creatively **interpreting the facts**, exaggerating one's position, and **promising more than one is currently able to deliver** are presented by some as clever manifestations of the entrepreneurial spirit.”

/Bucar, Branko and Robert D. Hisrich (2001)

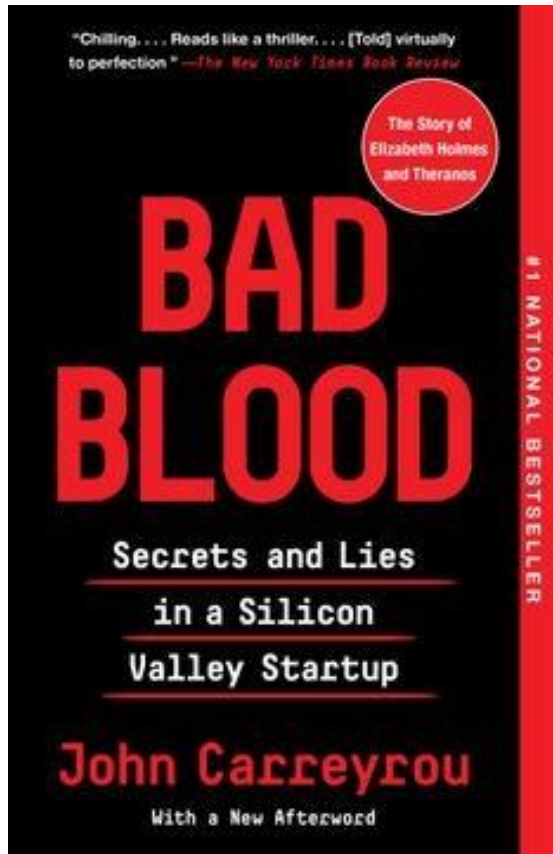
“When teaching how to pitch investors, for example, many entrepreneurship groups emphasize providing a polished vision of a product that may not in fact be ready.

This **“fake it till you make it”** approach that student entrepreneurship organizations teach **has its advantages,** allowing you to get a feel for user and investor interest, **but it also raises questions.**

When pitching investors, for example, **when does optimism cross the line into misrepresentation?** And how can startups hold onto core values while being open to investor and market feedback?”

Maurice Chiang / Stanford University

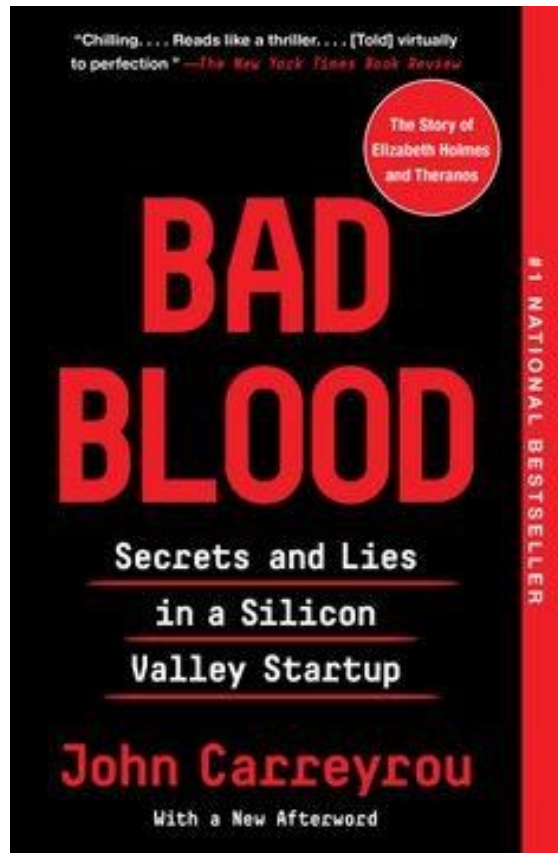
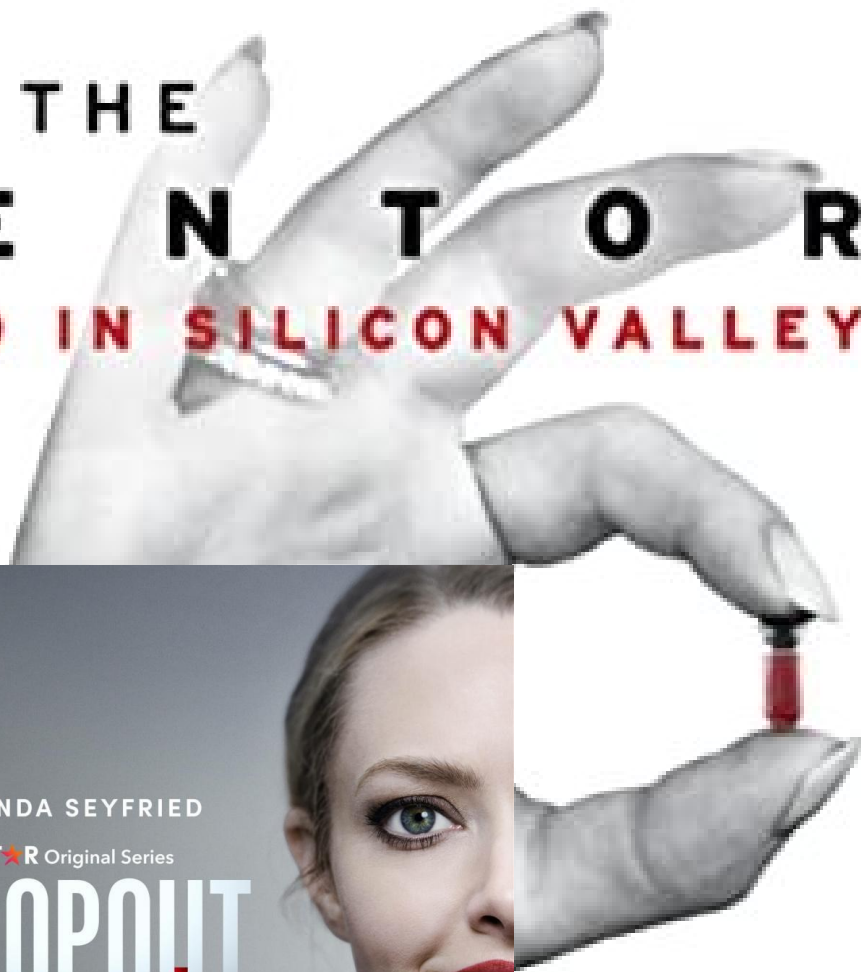
THE INVENTOR OUT FOR BLOOD IN SILICON VALLEY



HBO

<https://www.youtube.com/watch?v=Njqt88Gc6jl>

THE INVENTOR OUT FOR BLOOD IN SILICON VALLEY



HBO

THE FACT ABOUT THE FACEBOOK

YOU DON'T
GET TO
500 MILLION
FRIENDS
WITHOUT MAKING
A FEW
ENEMIES



JESSE EISENBERG

JUSTIN TIMBERLAKE

ROONEY MARA

the social network

A DAVID FINCHER FILM

DAILY COVER

Forbes

Marketing &
Advertising
GÖKCE
GÜVEN

30
UNDER
30

MEET THE CLASS OF 2025:
600 PIONEERS RESHAPING
THE FUTURE OF 20 INDUSTRIES—
AND THE WORLD

ET
NOW
ETNOW.IN



WORLD NEWS

26-year-old woman CEO and Forbes 30
Under 30 recipient Gokce Guven charged
with multiple counts of fraud

Questions for Early Stage Directors and Investors

As cases like Kalder continue to accumulate, boards and investors should be asking:

- What factual claims are we relying on when approving financings or public representations?
- How are customer, revenue, and technology claims independently verified?
- Are founders reusing investor materials for other regulatory or governmental purposes?
- Who has visibility into whether internal data aligns with external narratives?
- What mechanisms exist to surface concerns without retaliation or reputational fear?

ÉTICA RELACIONADA A LA TECNOLOGÍA

A promotional image for Shark Tank México. It features two women against a dark background with green and blue lighting. The woman on the left has long, wavy brown hair and is looking directly at the camera. The woman on the right has short brown hair, is wearing a bright pink jacket, and is holding a black phone to her ear. The text 'SHARK TANK MÉXICO' is in the top left, and the title 'LA CONTROVERSIA DEL BUSCADOR DE TAREAS' is in a green box across the middle.

**SHARK
TANK
MÉXICO**

LA CONTROVERSIA DEL BUSCADOR DE TAREAS



The Guardian - 2016



The New York Times - 2014

CLONE SHOCK



Ewe copycat... cloned sheep Dolly last night. Her birth has caused huge ethical storm

Sheep Dolly first mammal 'copied' from adult animal

A SHEEP has been cloned, scientists announced last night. The ewe, named Dolly, is the first mammal produced from an adult cell. She is named after busy country star Dolly Parton, because she was cloned from a mammary cell. But the breakthrough at Edinburgh's Roslin Institute has caused fury. It is feared if the technology is used on humans, a dictator could replace himself. That would echo the nightmarish movie *The Boys From Brazil*, where mini-Hitlers were created.

I LOOK LIKE EWE: PAGES 4 & 5

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I LOOK LIKE EWE: PAGES 4 & 5

TIME

FICTION BONUS:
"CLONE ON THE RANGE"
BY DOUGLAS COUPLAND

Will There Ever Be Another You?

A SPECIAL REPORT ON CLONING

10



Severance

Apple tv+

A person is seen from behind, in silhouette, looking at a wall covered in newspaper clippings and photos. The scene is dimly lit, with the person's face partially illuminated by light from the wall. The wall is covered in various newspaper clippings and photos, including a prominent photo of a woman in a hat in the top left corner. The overall atmosphere is one of investigation or discovery.

THE GREAT HACK



/the social dilemma

Ciencia Y Tecnología > Noticia



Justicia de EEUU culpa a Facebook, Instagram y YouTube de causar adicción y daños a adolescentes

La información fue dada a conocer por el Wall Street Journal, detallando que la justicia determinó que tanto Meta como Google fueron “negligentes al operar productos que dañan a niños y adolescentes, además de no informar sobre esos daños”.
Marzo 2026

Los padres de un adolescente que se quitó la vida demandan a OpenAI, creadora de ChatGPT



CORTESÍA DE LA FAMILIA RAINE

Nadine Yousif

BBC News

¿POR QUÉ ÉTICA ESPECÍFICAMENTE EN EMPRENDIMIENTO?

- Los emprendedores se enfrentarán a diferentes dilemas éticos que los no-emprendedores.
- Los emprendedores tendrán un fuerte impacto en la estabilidad económica del mundo.
- El emprendimiento innovador implica involucrarse con el status quo de las cosas. (e.g.: AirBNB y UBER con los problemas regulatorios y legales.)
- Los emprendedores se enfrentan a la expectativa de que la competencia es agresiva y a ciertos supuestos sobre cómo actúan los emprendedores (e.g.: *“los emprendedores tienden a doblar las reglas”*).

**¿A QUÉ DILEMAS ÉTICOS SE PUEDE VER
ENFRENTADO UN EMPRENDEDOR?**

Unavoidable Ethical Dilemmas for Entrepreneurs

[Home](#) › [Markkula Center for Applied Ethics](#) › [Focus Areas](#) › [Business Ethics](#) › [Business Ethics Resources](#) › Unavoidable Ethical Dilemmas for Entrepreneurs



The following list of ethical dilemmas, organized by the 14 stages of the Startup Lifecycle, are the product of the Markkula Center's Startup Ethics Online Community, The Ethical Startup, and roundtable discussions of prominent entrepreneurs and venture capitalists in the Markkula Center's network. Have a dilemma to add to the list? Head over to the Ethical Startup Group page at ethicalstartup.com to suggest it!

Business Ethics Resources

[A Model for Leadership Ethics](#)

[Create an Ethical Decision Making Framework for Your Organization](#)

[Articles](#)

[Cases](#)

[Event Summaries](#)

[Online Courses](#)

[Videos](#)

[Benison: The Practice of Ethical Leadership](#)

[Teaching Note: Interview of Theranos Whistleblower, Tyler Shultz](#)

ALGUNOS DILEMAS

- ¿Quién es **dueño de la idea**?
- ¿Dónde está la **línea entre exagerar y presentar bien tu proyecto**?
- ¿**Cuándo está listo el producto** para salir a mercado?
- ¿Qué **consecuencias tiene el desarrollo** de esta tecnología?
- A la empresa no le está yendo bien ¿**Cuándo es tiempo de renunciar**?

**¿CÓMO ENFRENTARSE A
UN DILEMA ÉTICO?**

ENSEÑAR ÉTICA EN EMPRENDIMIENTO



CONVERSANDO



HERRAMIENTAS

HERRAMIENTAS PARA ABORDAR EL RAZONAMIENTO ÉTICO Y LOS RIESGOS E IMPACTOS DE LA TECNOLOGÍA

8 Key Questions (8KQ's) Framework



The Madison Collaborative: “Ethical Reasoning in Action” is an educational initiative at James Madison University (JMU) that seeks to develop the ethical reasoning skills of the entire student body. The Madison Collaborative (MC) is specifically guided by the mission of “prepar[ing] enlightened citizens who apply ethical reasoning in their personal, professional, and civic lives.”

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ETHICAL OS 2018



ETHICAL OS 2021



A MODO GENERAL PREGUNTARSE

THINGS TO CONSIDER

- ❑ How could your product, service or business most **positively impact** the world you want to see? Beyond the problem you're trying to solve, how will the world be better for having this innovation introduced?
- ❑ Could your product or product feature harm someone directly? Consider not only the first-order consequences, but try to imagine the second- and third-order as well.
- ❑ Could a malicious actor use your product or product feature to harm a person or group? Commit a crime?
- ❑ Could the business model behind your product harm someone or a group of people? Could it fuel addiction? Could it impact certain groups—children, seniors, minority groups, the disenfranchised or disempowered—in different ways.
- ❑ Could your product risk someone's privacy? What happens if someone's data gets leaked, hacked, or lost? Do you have a plan in place for these contingencies? How can you best build trust with your users?
- ❑ Could your product or business do anything your users are unaware of? If so, why are you not sharing this information explicitly? Would it pose a risk to your business if it showed up in tomorrow's news?
- ❑ What can or should you do as a board to ensure your company lives up to the best version of itself?

For more details and the full Ethical OS toolkit:
ethicalOS.org

Are you making products?



THINGS TO KEEP IN MIND AS YOU GO

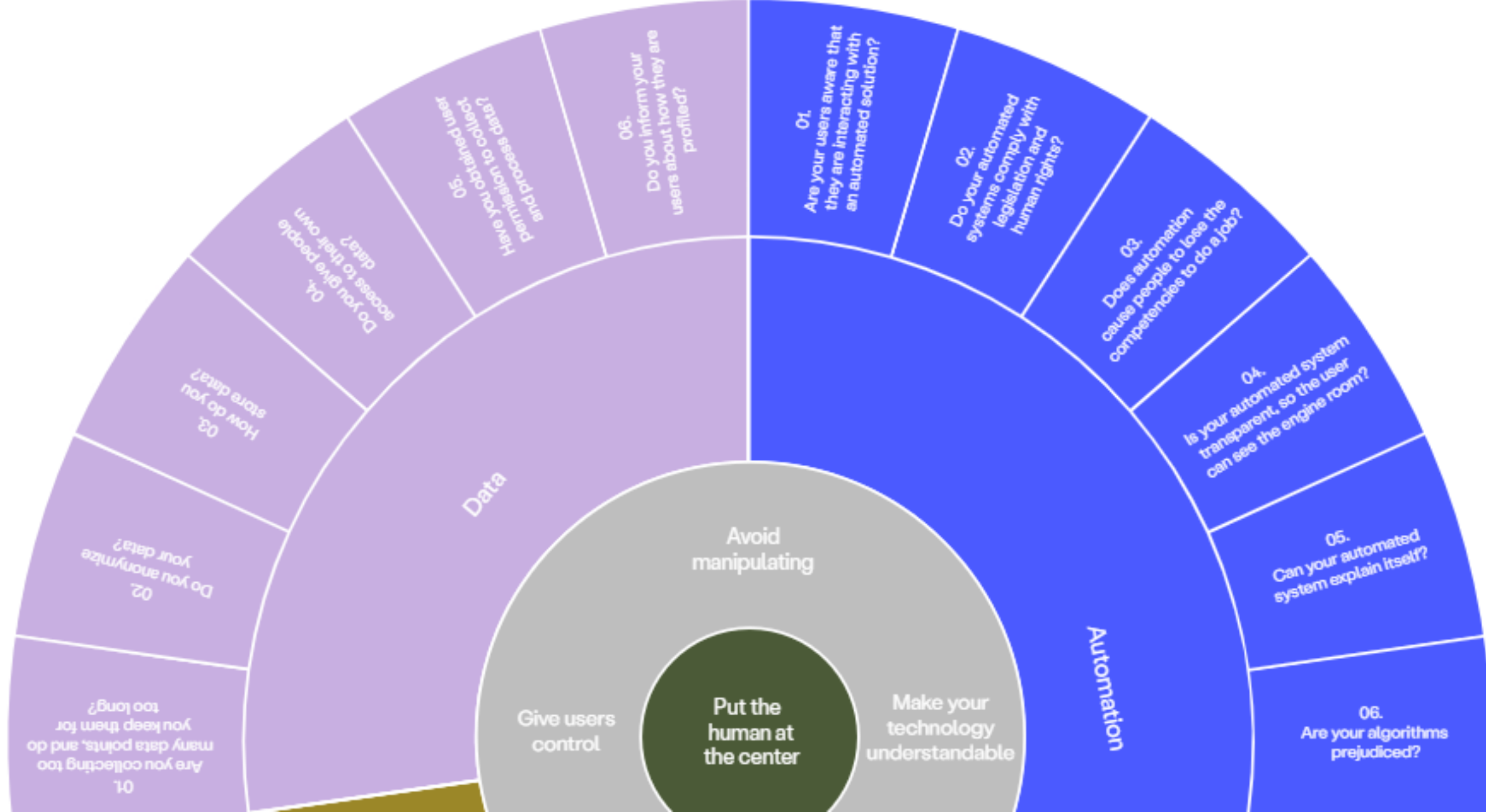
- ❑ Have you assessed not only the first-order consequences of your product, service or platform, but also its second- and third-order impacts?
- ❑ Have you designed your product or service in such a way so as to ensure that a diverse set of users will be able to access?
- ❑ Have you ensured you're collecting only as much user data as is absolutely necessary to operate your product and ensure your financial viability?
- ❑ Have you made certain that your training data or data sets represent a diverse set of users and that you've minimized any potential bias in the source of that data?
- ❑ Have you identified the most effective ways to ensure your product spreads only truthful content and information?
- ❑ Have you ensured the fairness, accountability, and transparency of any algorithms or machine learning processes embedded in your product or service?
- ❑ Have you designed appropriate off-ramps for your users? Have you eliminated, to the extent possible, bottomless pits, infinite scrolls, and attention traps?
- ❑ Have you found ways to avoid the use of any dark patterns in your UX?
- ❑ If you have them, are your notifications important enough to interrupt the user? Can they be delivered in a less intrusive manner—using, for example, a sound, a sensation, or subtle status change?
- ❑ Have you created systems that place a value on quality of user-generated content over quantity?
- ❑ Have you created contingency plans for data breach? Have you found ways to ensure that users will trust you before, during, and after such a breach?

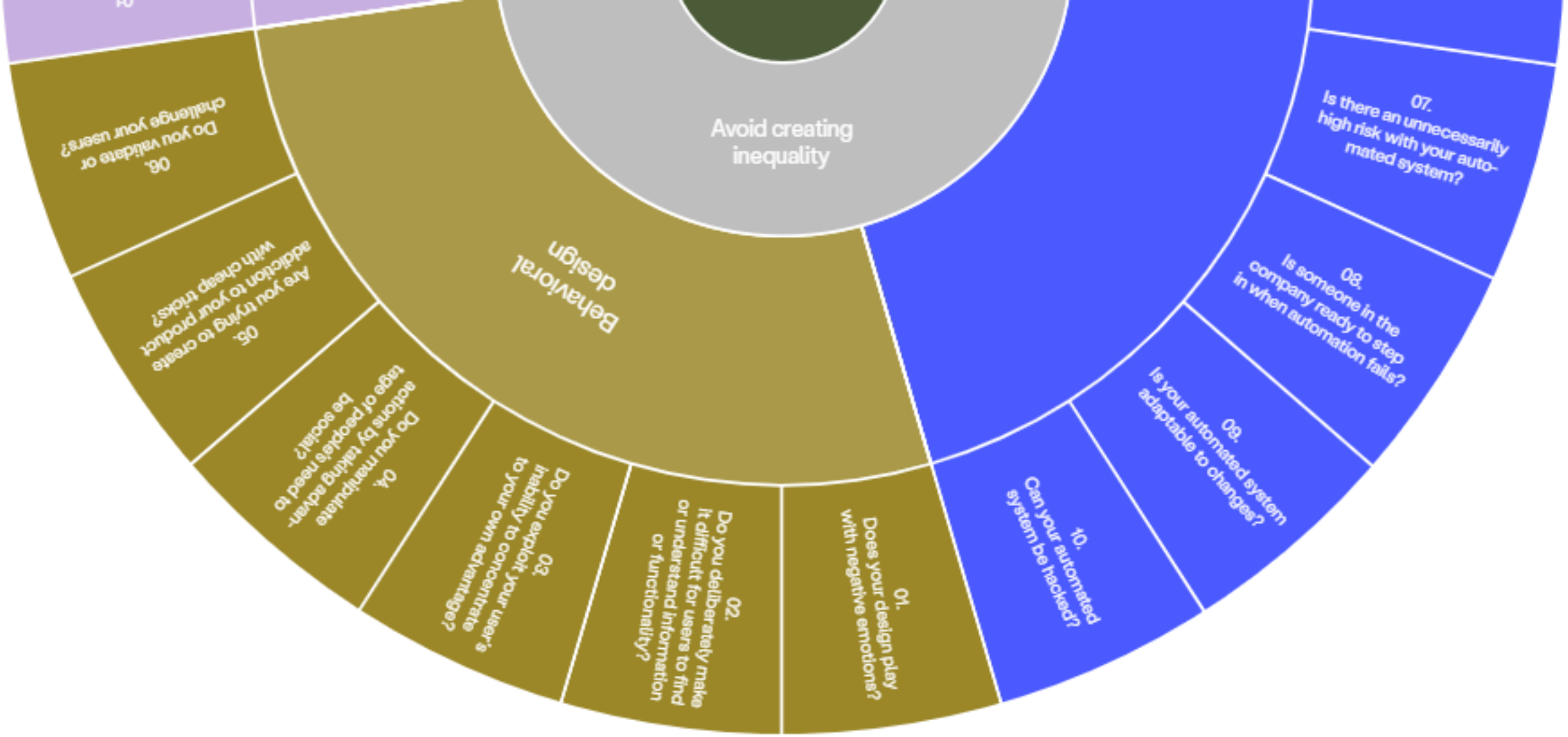
The Digital Ethics Compass

The compass is a tool to help companies make the right decisions from an ethical design standpoint.

The compass helps you ask the right ethical questions to avoid the most common design mistakes.







Pueden elegir una de las actividades, son similares.

ACTIVIDAD EN CLASES - Nuevas Perspectivas

1. Piensen en los stakeholders, en todas las personas que de una u otra manera son afectados por su solución. No sólo su público objetivo, sino también personas que incluso no lo usen pero igualmente les impacte como empleados, familiares, amigos de usuarios, etc. Van a tener tres tipos de stakeholders:
 - (i) Directos (clientes y usuarios que les afecta directamente)
 - (ii) Indirectos (no son usuarios pero les afecta)
 - (iii) Excluidos (que no pueden usar su solución por diversas razones)
2. Cada integrante del equipo identifique y describa un stakeholder distinto para su proyecto.
3. Ahora cada uno imagine una mala experiencia que haya tenido ese stakeholder relacionada a su solución y escriba una reseña a su producto/servicio personificando a ese personaje.
4. Conversen como equipo y descubran los desafíos éticos como una oportunidad de mejora.

N

BLACK MIRROR

ACTIVIDAD EN CLASES - “Provocatipo”

Diseña la versión menos ética y más irresponsable que se te ocurra para tu proyecto, un “provocatipo”, imaginando que no hay restricciones éticas ni límites, pero cumpliendo con que tu solución debe ser usable y solicitada por sus usuarios.

1. Describe la solución original en dos líneas
2. Describe la nueva versión no ética en 3-5 líneas.
3. Por qué esa versión es poco ética? A quién daña y por qué?
4. Qué parte de la solución original inspiró la versión no ética?
5. Podría tu solución transformarse, aunque sea un poco, en su versión no ética.



¿Podría tu solución ser parte de un capítulo de Black Mirror? Imagina el plot.

ACTIVIDAD EN CLASES - Preguntas para conversar

Conversen las siguientes preguntas según el número que saquen

1. Tu solución representa la persona que eres? Por qué?
2. Tu solución representa el futuro que quisieras ver? Por qué?
3. Tu solución aumenta desigualdades? Por qué?
4. Tu solución hace que las personas actúen de una forma que de otra manera no lo harían?
Es un problema? Es fácil entender los mecanismos profundos de tu solución? Por qué?
5. Te has imaginado a ti mismo como usuario de tu solución o afectado por ella? Sería una buena o mala experiencia?
6. Qué es lo peor que puede pasar con tu solución?
7. Qué argumentos en contra de tu solución podría decir un fuerte opositor?
8. Que pequeños cambios podrían mejorar, desde el punto de vista ético, tu solución?
9. Las personas adquieren mayor o menor control de sus vidas como resultado de tu solución? Es bueno o malo?
10. Desafías o mantienes el status quo con tu solución? Es bueno o malo?
11. Estaría mal que los principios de tu solución se transformaran en principios generales?

CURSO ING2030

INVESTIGACIÓN, INNOVACIÓN Y EMPRENDIMIENTO

ANEXOS

HERRAMIENTAS EN DETALLE

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What do legitimate authorities (e.g., experts, law, my religion/god) expect of me?

8. RIGHTS

What rights, if any, (e.g., innate, legal, social) apply?

8 KEY QUESTIONS (8KQ) FRAMEWORK

1. FAIRNESS

What action does justice, equality, or a balance of legitimate interests require?

What is the fair or just thing to do?

How can I act equitably and treat others equally?

A basic strategy when asking about fairness is to begin with intuitions of fairness, and critically examine them to ensure that we are seeing past any self-interest to actually treat everyone including ourselves equitably. In doing so, the “veil of ignorance” is a useful analytical device.

8 KEY QUESTIONS (8KQ) FRAMEWORK

2. OUTCOMES

What achieves the best short- and long-term outcomes for me and all others?

A basic strategy is to think about all those involved in the situation, currently living and yet to be born, who will be helped or harmed by an action, and to total the overall positives and negatives. This is an exceedingly difficult process to not play favorites with one's self or loved ones, but also to forecast whether an action will be a net positive or negative. In some dilemma situations the best that one can do may be to minimize the negative outcomes.

8 KEY QUESTIONS (8KQ) FRAMEWORK

3. RESPONSIBILITY

What duties and/or obligations apply?

A basic strategy here that can focus thoughts about responsibilities is (1) to ask if the persons associated with the ethically significant situation have any role-based responsibilities or owe any natural duties; if so, to clarify what those are; and (2) to ask if those responsibilities are outweighed by other ethical factors.

8 KEY QUESTIONS (8KQ) FRAMEWORK

4. CHARACTER

What action best reflects who I am and the person I want to become?

What actions will help me to become my ideal self?

What action expresses the person that I am or aspire to be?

What virtues, e.g. honesty, courage, integrity, loyalty, do I want to nurture?

What vices do I want to eliminate?

8 KEY QUESTIONS (8KQ) FRAMEWORK

4. CHARACTER

A basic strategy is to begin by a period of personal reflection to help an individual decide for herself what ideal self she aspires to become. This leads to the identification of a list of virtues, or traits of character, that would be associated with that ideal. When faced with an ethical decision, one should act in accordance with these identified traits of character. Deciding on the best action involves testing each virtue – What would an honest person do? What would a kind person do? And so on.

8 KEY QUESTIONS (8KQ) FRAMEWORK

5. LIBERTY

How does respect for freedom, personal autonomy, or consent apply?

A basic strategy here that can focus thoughts about liberty is to think about whether the people associated with the morally significant case are entitled to live as they see fit. If they are not hurting others in any obvious ways through their choices and actions, and they are capable of making rational choices about their own self-determination, there is a strong presumption in favor of their being allowed to continue doing so. One should act so as to increase personal liberty and autonomy unless the persons involved are incapable of living out their autonomy or are harming others.

8 KEY QUESTIONS (8KQ) FRAMEWORK

6. EMPATHY

What would I do if I cared deeply about those involved?

A basic strategy here is (1) to imaginatively put yourself in the place of those who would be affected by your action/inaction (or the action/inaction of others), and then (2) to see how you would feel about that action/inaction from his/her/their perspective. If you would feel disrespected or otherwise harmed by the action/inaction, there is a presumption in favor of not doing it (or judging it acceptable).

8 KEY QUESTIONS (8KQ) FRAMEWORK

7. AUTHORITY

What do legitimate authorities (e.g., experts, law, my religion/god) expect of me?

A basic strategy here is to question (1) the basis for an apparent authority figure or institution's claim to authority, and seek to explain the justifications for that authority, i.e. determine legitimacy, (2) whether the advice or commands are consistent with the other seven key questions, and (3) whether consciously breaking the proscriptions of authority, even if legitimate, is required by the ethical demands of the other questions.

8 KEY QUESTIONS (8KQ) FRAMEWORK

8. RIGHTS

What rights, if any, (e.g., innate, legal, social) apply?

A basic strategy for focusing thoughts about rights is (1) to think about whether there are any basic entitlements and permissions necessary for living a life with dignity at stake in the situation, (2) to consider whether there are any instrumentally valuable protections or permissions established by legitimate authorities such as law or government, and (3) to ask if respecting these rights is outweighed by competing considerations (outcomes, responsibilities, etc.). Rights create a strong (for some inviolable) presumption for being respected.

THE ETHICAL OPERATING SYSTEM (ETHICAL OS) 2021

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ETHICAL OS



ETHICAL OS

1

LAW ENFORCEMENT

Widespread support of the Black Lives Matter movement that grew during the summer of 2020, along with increased scrutiny of police tactics and surveillance technology, has brought the issue of ethics and fairness in law enforcement into an urgent public conversation. Recent reports highlight the dangers of predictive policing systems, including a controversial one in Pasco County, Florida by which an algorithm created a pool of “prolific offenders” who have subsequently reported continued harassment. Calls to defund the police, in part by rerouting emergency calls to social workers and others trained in de-escalation, have raised the question why there have been armed officers responding to non-violent situations in the first place. Accusations of systemic racism within law enforcement have been named as a reason for vastly unequal treatment of Black and Brown residents, and for the disproportionate killings of Black people at the hands of the police.

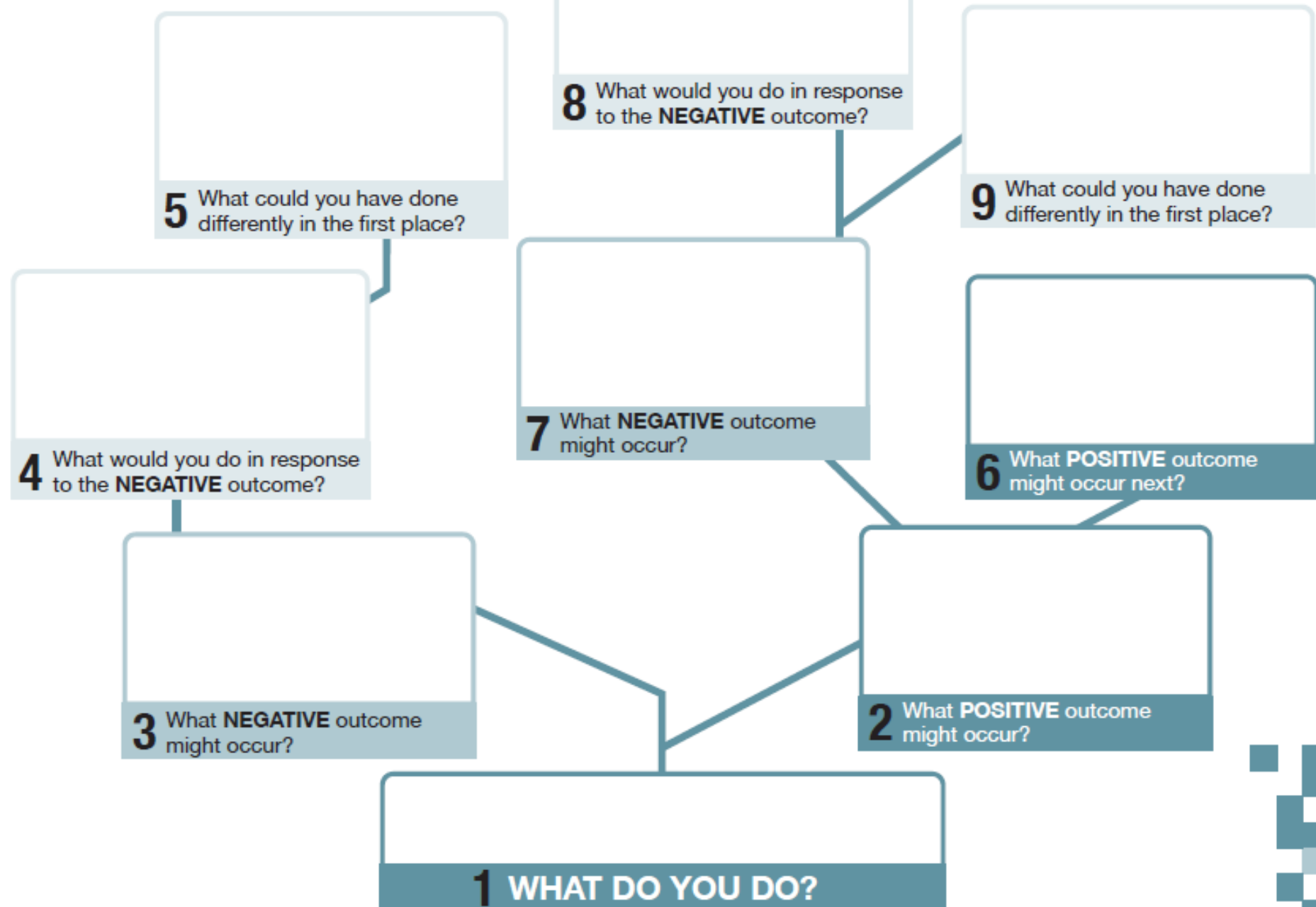


2022: Algorithmically Accused

Your county recently began using a facial recognition system to identify individuals caught on surveillance cameras. Recently, on the recommendation of this system, your department arrested a Black resident at his place of work, drove him to a detention center, and interrogated him for several hours before determining the system had misidentified him.

Activist groups are calling for the county to cease using the software immediately, claiming the system violates resident privacy and is inherently biased against non-white populations. These groups are lobbying to ban every police force in the state from using any facial recognition technology.

To counter the movement to ban this technology, you have been tasked with proposing publicly available, legally enforceable guidelines on the acquisition, implementation, and oversight of the algorithms that facial recognition systems use when making criminal justice decisions in your county.



2022: Decision Tree

LAW ENFORCEMENT
Algorithmically Accused



? What can be done by the county or the state to determine the fairness of an AI algorithm sold to them by a private firm?

? How would you go about building a team to evaluate risks and implementation of algorithm-based technologies?

? Under what conditions should AI models that identify potential suspects be banned in civil and criminal investigations?

? If information about potential suspects is gleaned from these AI models, does it need context and caveats to be considered by a jury or judge?

? Should the government take a more active role in structuring and/or regulating the AI algorithm that makes these determinations, and if so, how might this be done?

? Where do you see similarities to your own work where you've had to make decisions that balance fairness and safety? Does using the Decision Tree change how you might have done things or how you will going forward?



2032: Embedded Governance

Embedding penal codes into public spaces through networked sensors has transformed law enforcement. Cities can download laws directly into the built environment. Automated inspections issue and collect payments for taxes and fines, reducing corruption and mismanagement. Roads issue speeding tickets, reducing officer-involved shootings ten-fold.

A new technology has appeared that tracks physical movements, analyzes written and spoken public expressions, and follows purchasing patterns to determine if someone is a threat to themselves or others. For residents flagged as threats, a series of interventions are deployed: disabling their car remotely, locking them out of registered biometric firearms, and sending capture drones to any school, church, or large gathering they enter. You are considering implementing this technology, which was used to thwart a potential mass killing in Vancouver last Christmas, even over the objections of privacy groups.

2032: Decision Tree

LAW ENFORCEMENT
Embedded Governance



ETHICAL OS

2

PUBLIC HEALTH

Public health will continue to emerge as a crucial focus area for governments over the next decade. It will touch and be touched by issues spanning multiple sectors, demographics, and geographies, with disparate outcomes among different populations. Governments will be accused of both overreach and inaction, and will be forced to dig deep to address the systemic root causes of public health inequities. Technological advances will exacerbate public health crises, such as social media's impact on mental health and the amplification of misinformation. Policymakers will need to consider the long-term impacts of technology on the public's ability to manage its physical, mental, and social health.



2022: Everyone's a Therapist

Exacerbated by an unending pandemic, social unrest, and high unemployment, mental health has deteriorated across the city. Drug overdoses are increasing. Depression rates are rising, even in kids as young as 10 years old.

All departments and city employees are being asked to prioritize mental health, regardless of their role or ability, reminiscent of when each department was asked to determine how it could best contribute to the city's climate goals. However, the historical stigma associated with mental health means people are reluctant to discuss the issue.

Your department has been given a list of technology interventions to implement—from wearables for city employees to sensors in schools that can identify pre-aggressive and depressive behavior—to address the growing mental health crisis. They're costly, and your staff has raised privacy concerns, worried that information collected might hurt their career advancement, but you aren't given much choice. You've also been asked to create guidelines for any job roles and responsibilities that might pose a risk to city employees' mental health.



2032: The Next Pandemic

When the Pandemic Early Warning System alerted the world that COVID-32 was on its way, your team quickly activated the state government's predictive monitoring systems, embedded in all phones, cars, buildings, and public transit.

Just as quickly, anti-science factions mobilized a campaign to discredit all public health messaging. They deployed surveillance scrambling technology, hampering your ability to set up tracking and treatment infrastructure, and filed a lawsuit claiming you'd violated the state's Freedom From Technology statute.

A large social media company has offered to help work around these groups by disseminating public health information, cracking down on misinformation, and sharing tracking, purchase, and movement data with you. In exchange, they want to be the state's primary health care service provider.

Their solution seems to be the best option, but you're concerned they're taking advantage of the emergency to collect information from residents who won't be able to opt out.

ETHICAL OS

3

EQUITY AND INCLUSION

From the streets to the boardroom to the chambers of City Hall, a movement demanding equity has become mainstream. More and more departments will be tasked with implementing an equity lens across their programs and services, and they will be mandated to achieve equitable outcomes in everything from health to income to education to transportation. Equity can be hard to define because it is not just an outcome, rather it is a process: one of examining implicit biases and default designs in new technologies, as well as the ways they exist in legacy systems and structures. Equity is being embedded into every aspect of governance, seen in the creation of Oakland's Department of Race and Equity, Austin's Equity Office, and Portland's Office of Equity and Human Rights, to name a few. Governments will have to consider how each new program, service, or offering will move towards or away from an equitable future.



2022: Universal Basic Reparations

Your county has committed a monthly payment to individuals negatively affected by racial injustice. You've set up a website for people to register themselves to receive this payment and have sent workers to canvass city neighborhoods to ensure everyone who is eligible signs up.

50% of people who sign up do so via direct deposit, and their payments are sent without hassle. But five months into the program, the unbanked population has yet to receive a single payment. Advocates have been urging you to allow a direct cash transfer or mail a pre-paid card instead of a check to avoid high check-cashing fees, but county regulations don't allow it. Your higher-ups are considering other approaches such as loan forgiveness, college investment funds, or other benefits in lieu of cash payments to ensure funds are used "wisely and appropriately." But you're convinced that people can be trusted to use unrestricted funds in a way that would be most in service of their needs.

ETHICAL OS

4

ARTIFICIAL INTELLIGENCE

As AI and machine learning get exponentially smarter and more ingrained into our lives, governments will shift resources to funding, regulating, and employing AI assistance in everything from law enforcement to health monitoring to food systems, such as a new USDA-funded institute focusing on using AI to optimize our food systems. More accurate predictive modeling can, in theory, lead to better governing outcomes across the board, but these new technologies are more often than not built with algorithmic biases that will need to be addressed. Concerns of privacy, security, and government overreach are also deeply embedded within AI-integrated governing technologies.



2022: AI for Individualized Marketing to Residents

The new mayor of your city recently overhauled your communications department to use an AI analysis of individual resident preferences, resulting in highly targeted, individualized messaging designed to achieve a wide range of policy outcomes. The utility company sends ads about electrical vehicle rebates to households they think will be in the market for a new car. The health department blocks “Restaurant Week” announcements from post-pandemic anti-mask advocates. Many city departments have begun experimenting with new ways to use this system to encourage (and restrict) resident activities. However, a small but growing number of people are becoming increasingly concerned the city has crossed a line from policing criminal behavior to trying to control all behavior. It’s a fine line, they say, between the government pushing values and pushing products.

In response, you’ve been asked to develop a set of public guidelines for departments as they make use of this new communications tool.

ETHICAL OS

5

CLIMATE

Addressing climate change is the issue of our time. While growth in sustainable materials, emissions monitoring, and adaptation techniques are advancing, we cannot engineer our way out of climate change. Similar to equity, governments will need to take a climate lens to everything they do. Recent examples include President Biden's executive order to embed climate policy across nearly every Federal agency, or Birmingham, Alabama's comprehensive plan for sustainability policies written into every department. Departments and agencies that have not historically been part of the climate conversation will need to learn how to incorporate sustainability priorities into their operations. Whether at an individual, industrial, or institutional level, it will become everyone's responsibility to address the impacts of violent climate events and to collaborate on collective actions to reduce our carbon footprint on the earth.



2032: Citizen and Robot Science

Using specimen collection bots and machine-learning drones, a citizen scientist organization recently open-sourced a pollution map of your city, displaying pollution levels alongside other data like traffic levels, building construction, and health outbreaks.

This group also created an early warning system, sending drones to swarm neighborhoods to alert residents to stay inside when pollution levels rise, based on real-time analyses of collected data. Last week, the drones swarmed a neighborhood whose residents had been required to sign a document acknowledging an increased risk of environmental exposure prior to moving there. This contract included a provision guaranteeing residents would be continually updated by the city about elevated risk levels.

Since the drone alerted them before the city did, residents are angry at the city for failing to meet its commitment to inform residents of potential exposure. The Department of Public Health is now proposing a lengthy regulatory process to operate the drones, but the organic system is very popular among the public. As a public liaison, you worry regulating them will further decrease the public's already fading trust.

A MODO GENERAL PREGUNTARSE

THINGS TO CONSIDER

- ❑ How could your product, service or business most **positively impact** the world you want to see? Beyond the problem you're trying to solve, how will the world be better for having this innovation introduced?
- ❑ Could your product or product feature harm someone directly? Consider not only the first-order consequences, but try to imagine the second- and third-order as well.
- ❑ Could a malicious actor use your product or product feature to harm a person or group? Commit a crime?
- ❑ Could the business model behind your product harm someone or a group of people? Could it fuel addiction? Could it impact certain groups—children, seniors, minority groups, the disenfranchised or disempowered—in different ways.
- ❑ Could your product risk someone's privacy? What happens if someone's data gets leaked, hacked, or lost? Do you have a plan in place for these contingencies? How can you best build trust with your users?
- ❑ Could your product or business do anything your users are unaware of? If so, why are you not sharing this information explicitly? Would it pose a risk to your business if it showed up in tomorrow's news?
- ❑ What can or should you do as a board to ensure your company lives up to the best version of itself?

For more details and the full Ethical OS toolkit:
ethicalOS.org

Are you making products?



THINGS TO KEEP IN MIND AS YOU GO

- ❑ Have you assessed not only the first-order consequences of your product, service or platform, but also its second- and third-order impacts?
- ❑ Have you designed your product or service in such a way so as to ensure that a diverse set of users will be able to access?
- ❑ Have you ensured you're collecting only as much user data as is absolutely necessary to operate your product and ensure your financial viability?
- ❑ Have you made certain that your training data or data sets represent a diverse set of users and that you've minimized any potential bias in the source of that data?
- ❑ Have you identified the most effective ways to ensure your product spreads only truthful content and information?
- ❑ Have you ensured the fairness, accountability, and transparency of any algorithms or machine learning processes embedded in your product or service?
- ❑ Have you designed appropriate off-ramps for your users? Have you eliminated, to the extent possible, bottomless pits, infinite scrolls, and attention traps?
- ❑ Have you found ways to avoid the use of any dark patterns in your UX?
- ❑ If you have them, are your notifications important enough to interrupt the user? Can they be delivered in a less intrusive manner—using, for example, a sound, a sensation, or subtle status change?
- ❑ Have you created systems that place a value on quality of user-generated content over quantity?
- ❑ Have you created contingency plans for data breach? Have you found ways to ensure that users will trust you before, during, and after such a breach?

**THE ETHICAL OPERATING SYSTEM
(ETHICAL OS)
VERSIÓN ANTIGUA 2018**

ETHICAL OS



ETHICAL OS

1

TRUTH, DESINFORMATION, PROPAGANDA

Shared facts are under attack. By everything, from fake news to bots that spread propaganda while posing as real people. Now a new wave of dangerous disinformation is on the rise. Known as “deep fakes,” these highly convincing videos algorithmically alter or replace people’s speech, facial expressions, and identities to create fake proof of actions or speech that never actually happened. Awesome.

Many individuals and groups are highly motivated to subvert the truth at a massive scale, especially for political ends. New social media technologies will make it even easier to spread lies and undermine trust.

Over the next decade, what else could be faked via new technologies? Which shared truths, facts, and information will we be called on to protect?

ETHICAL OS

2 ADDICTION & THE DOPAMINE ECONOMY

Research by Common Sense Media found that the average teenager spends 9 hours a day using some form of media. 9 hours!

The time we spend with our devices is of growing concern. Tristan Harris, founder of the Center for Humane Technology (CHT), has called for tech companies to encourage “time well spent,” suggesting designers optimize the time they spend on platforms, in a way that makes their time beneficial to their overall happiness and well-being.

Studies show people achieve maximal intended use of apps like Instagram and Snapchat after 11 minutes—beyond which overall happiness decreases. How might tools be designed to advocate for time well spent? How can we design software that prioritizes user happiness—offline and online—over keeping eyes glued to the screen?

ETHICAL OS

3

ECONOMIC & ASSET INEQUALITIES

In 2017, according to Oxfam International, eight people owned as much wealth as the entire bottom half of the world's population. Let that sink in a minute.

Wealth concentration and distribution have been issues issue throughout modern history. But it's gotten way worse. Today, in the United States, wealth concentration is the worst it's been since 1928. New technology can democratize access, provide income opportunity, and balance distribution, but it can also exacerbate inequality by catering only to high-income groups and eliminating low-income jobs.

ETHICAL OS

4 MACHINE ETHICS & ALGORITHMIC BIASES

As the application of AI in critical domains like welfare, education, employment, and criminal justice has intensified, so, too, have concerns around the way AI can reflect, introduce, or even amplify bias.

Whether it's racially biased police facial recognition systems or search engines that privilege certain ideas and identities, there is enough evidence to know that technology itself is not neutral. That is because technology is often the product of human action or a tool used by humans—humans who by nature are not neutral.

Solutions that rely on AI to remedy challenges like those above often overlook the fact that AI itself is a tool created and used by humans. An interdisciplinary approach to the design, use, and governance of AI technology remains critical to addressing some of the inadvertent and/or intentional discriminatory impacts on certain individuals and communities.

ETHICAL OS

5

SURVEILLANCE STATE

Recent examples of social bots being co-opted by governments and militaries for use in attacking the opposition reveal that technology built for benign uses can be leveraged for harm. In some cases, armies of automated software-driven Twitter and Facebook profiles were used to target journalists, activists, and citizens who spoke out against particular ideas.

In authoritarian countries, Western surveillance tools have powered state sponsored repression, allowing leaders to intercept emails and text messages and monitor the whereabouts of citizens through their phones. Armed with this information, police in many of these countries now routinely confront dissidents with records of their messages and movements during arrest and torture.

Meanwhile, increasing use of facial recognition and social media tracking is allowing governments to create deep records of individual behavior. “Citizen scores” generated over years of data collection can be used to limit access to public spaces, work opportunities, and more, creating a new kind of social inequality based on prior behavior and speech.

ETHICAL OS

6

DATA CONTROL & MONETIZATION

In the future, users will expect access to tools for acquiring, sharing, interpreting, and verifying information that has been collected about themselves and their environment. They will assert a fundamental right to selectively protect the privacy of their personal and institutional data where disclosure would jeopardize their well-being. And they will expect to control access to their personal data and have opportunities to selectively share, monetize, and benefit from it alongside the tech companies that create and use the data.

ETHICAL OS

7 IMPLICIT TRUST & USER UNDERSTANDING

Misuse of data is a serious problem. But perhaps equally as troubling is the fact that many users are not sufficiently aware of how exactly popular apps and platforms work, how their engagement is optimized, or what's being tracked and collected. Terms of service are hardly read and are rarely written for clarity. Permissions to many activities are often given without full understanding of how they will be used.

In the future, companies are likely to face backlash when their product or employees violate the implicit or explicit assumptions and scope of what users have agreed to, especially where that agreement is based on users accepting a dense, indecipherable agreement no one actually reads.

ETHICAL OS

8

HATEFUL & CRIMINAL ACTORS

On August 12, 2017, in Charlottesville, VA, Heather Heyer was killed when a man linked to a white supremacist group, crashed his car into a crowd of counter protestors at a rally called “Unite the Right.”

The rally was organized using the Facebook events tool.

Obviously, hate crimes are nothing new. But online tools enable the global dissemination of content with an ease and speed never before available to those wishing to spread hate. Bullying, radicalization, trolling, doxing and other bad behaviors have boomed, thanks to tech. And it’s practically impossible to mitigate or prevent negative experiences for users who spend a great deal of time navigating in that environment.

THE DIGITAL ETHICS COMPASS

The Digital Ethics Compass

The compass is a tool to help companies make the right decisions from an ethical design standpoint.

The compass helps you ask the right ethical questions to avoid the most common design mistakes.



